

SSAA–Texture-Oversampling Resonance, Convergence Plateaux, and Implications for DIC-UQ Image Rendering

1 Purpose and terminology

This note records the interpretation of the convergence figures used in the Exp1 and Exp2 rendering studies. “SSAA” (S) denotes the number of regular camera samples along one final-image pixel axis. “Texture oversampling” (O) denotes the number of stored texture texels along the corresponding axis of one final-image pixel. Thus a setting (S, O) has S^2 camera samples and O^2 texture texels per final pixel.

The central distinction is between two limits:

1. *raster convergence*: increasing S for a fixed stored texture;
2. *texture convergence*: increasing O towards a latent, continuous surface-intensity field.

Increasing one does not recover error due to the other.

2 The matched-grid dip at $S = O$

The texture-generation pipeline stores area averages of the continuous pattern over texture texels. At an undeformed, grid-aligned frame, $S = O$ partitions a camera pixel into precisely the same cells. With a cardinal interpolant such as Catmull–Rom evaluated at texel centres, averaging those stored texel averages reproduces the camera-pixel area average. This produces the very low, and at frame zero often machine-precision, diagonal trough visible in the figures.

It is a matched quadrature/reconstruction condition, not evidence that a general continuous texture reconstruction has converged. It is weakened after deformation because sample points no longer remain aligned with the original texture-cell lattice. It is especially fragile for discontinuous or nearly discontinuous content, such as the edges of additive disks. No renderer branch special-cases $S = O$; the behaviour follows from the common grids and preintegrated texture texels.

The diagonal is therefore useful as a regression test for UV alignment, texture-centre conventions, and frame-zero agreement, but should not be used as a general production-convergence criterion.

3 Where the plateaux lie

For a fixed oversampling O , increasing S crosses the matched-grid diagonal as follows:

$$S < O \quad \longrightarrow \quad S = O \text{ (possible resonant dip)} \quad \longrightarrow \quad S > O \text{ (raster-convergence plateau)}.$$

The post-diagonal plateau, if it has been reached, is the error of rendering a fixed reconstructed texture; more camera samples cannot remove its intrinsic texture/interpolation error. It is the relevant fixed-texture raster limit.

Conversely, at fixed S , an oversampling sweep crosses the diagonal in the opposite direction. Its post-diagonal side has $O > S$ and is useful for testing whether improving texture resolution

lowers the texture-error floor. These are different one-dimensional paths through the same two-parameter error surface.

For smooth Gaussian speckles, there are examples in the present affine, frame-two data where the $O > S$ side is both stable and sub-LSB. For example, at $S = 16$, the maximum floating-point error at $O = 32, 64, 128$ is approximately 4.9×10^{-5} , 6.1×10^{-5} , and 6.4×10^{-5} , respectively: a stable value below one 12-bit LSB ($1/(2^{12} - 1) \simeq 2.44 \times 10^{-4}$). At $S = 32$, the corresponding $O = 64, 128$ values are approximately 1.2×10^{-5} and 1.5×10^{-5} , at the one-16-bit-LSB scale.

For additive disks, the analogous post-dip limit can be much larger. For example, for affine frame two and $O = 16$, the fixed-texture values at $S = 64, 128$ are approximately 9.8×10^{-3} and 9.6×10^{-3} . This is a clear plateau on the $S > O$ side, but it is not sub-LSB even at 8 bits. Other disk combinations need more post-diagonal SSAA samples before a plateau can be claimed; passing $S = O$ alone is insufficient.

4 Consequences for texture choice and render convergence

The ideal Gaussian pattern is a deliberately favourable smooth test. It reduces high-frequency content and makes both texture and raster convergence easier, but does not make $O = 1$ intrinsically exact. An additive disk is a useful sharp-edge stress case. Real printed or painted patterns are more closely represented at the specimen surface by irregular, finite-edge disks with a distribution of radii and contrast, rather than by either ideal limit.

For a robust convergence claim, choose O sufficiently high that the texture limit is acceptable, then increase S beyond O and demonstrate a stable fixed-texture plateau. The $S = O$ point should be reported separately or visibly identified as a matched-grid condition. Sharp textures require higher O and may require higher S to establish the plateau; a high S cannot recreate edge information missing from a low- O texture.

5 Implications for conventional DIC-UQ workflows

In common DIC-UQ practice, a captured reference camera image is used directly as the texture. Relative to that camera grid it is an $O = 1$ texture. The input image already contains the effects of the reference optical point-spread function (PSF), pixel integration, sampling, noise, and quantisation. On deformation, an interpolant can only impose an assumed continuous reconstruction of those samples; increased target-render SSAA converges that assumption, not the unknown latent surface pattern.

Consequently, two renderer modes should remain distinct:

Mode	Input texture	Target-camera PSF
Latent-scene truth	pre-camera high-resolution pattern	applied once
Camera-image texture	captured $O = 1$ reference image	not applied again

Applying a PSF again in the camera-image-texture branch would double blur the reference, including at frame zero. Moreover, image interpolation does not physically transport the baked-in reference PSF. Under affine or nonlinear deformation, warping the reference image warps its effective PSF and pixel footprint, whereas the target camera PSF is fixed in target-camera coordinates. The approximation is comparatively benign for translation, but is not exact for strain, scaling, shear, or spatially varying deformation.

An effective image-plane PSF is therefore a useful, bounded sensitivity study: apply it once in the latent-scene truth branch before final pixel integration, and compare that truth with the no-second-PSF $O = 1$ camera-image workflow. A PSF/defocus attenuates troublesome high frequencies and can make the standard workflow accurate enough, but it does not guarantee correctness. The outcome depends on the effective camera MTF, speckle scale, pixel pitch,

interpolation, and maximum local deformation. The practical result to seek is a tolerance map: the combinations of effective PSF width and deformation for which the $O = 1$ workflow stays below the selected image or DIC-UQ error tolerance.

6 Recommended reporting

The main study can retain the no-PSF sharp-texture cases as transparent upper-bound stress tests. For practical relevance, add a small, explicitly labelled effective-PSF sensitivity section rather than a full optical model. Report: (i) the $O = 1$ camera-image-texture error relative to a latent-scene reference, (ii) the fixed-texture $S > O$ plateau, (iii) the separate role of texture oversampling, and (iv) the effect of one or a few plausible PSF widths. This directly answers when image-as-texture DIC-UQ is adequate and when it is not.